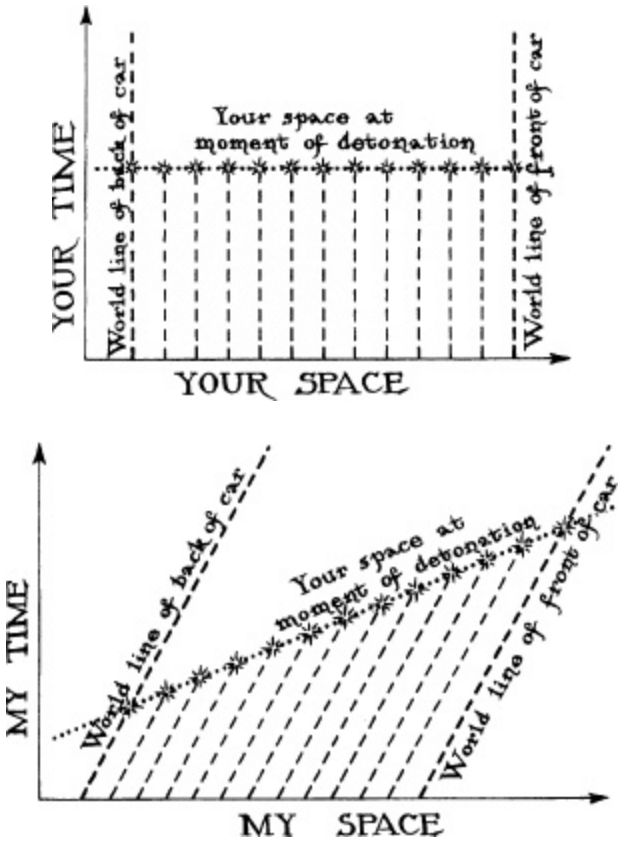



From Figure 1.3c it is clear that, to move through your space at your moment of detonation (along the dotted detonation line), I must move through both my space and my time. In this sense, your space is a mixture of my space and my time. This is just the same sense as the statement that magnetic north is a mixture of true north and true east (compare Figure 1.3c with Figure 1.2).

You might be tempted to assert that this “mixing of space and time” is nothing but a complicated, jingoistic way of saying that “simultaneity depends on one’s state of motion.” True. However, physicists, building on Einstein’s foundations, have found this way of thinking to be powerful. It has helped them to decipher Einstein’s legacy (his new laws of physics), and to discover in that legacy a set of seemingly outrageous phenomena: black holes, wormholes, singularities, time warps, and time machines.

From the absoluteness of the speed of light and the principle of relativity, Einstein deduced other remarkable features of space and time. In the language of the above story:

- Einstein deduced that, as you speed eastward down Colorado Boulevard, I must see your space and everything at rest in it (your